

Gas Laws

The three variables are connected. The trick is always: identify what is held constant, then choose the law.

FOUNDATION

Ideal gas picture

- Particles have negligible volume compared with container.
- No intermolecular forces except during brief elastic collisions.
- Random straight-line motion between collisions.

State variables

- p : pressure from particle collisions with walls.
- V : volume occupied.
- T : related to average kinetic energy.
- n : amount of gas.

THE THREE GAS LAWS

Boyle-Mariotte · T constant

- $p \propto 1/V$ — compressing the gas raises pressure.
- $p_1 V_1 = p_2 V_2$.

Gay-Lussac · V constant

- $p \propto T$ — temperature must be in Kelvin.
- $p_1/T_1 = p_2/T_2$.

Charles · p constant

- $V \propto T$ — temperature must be in Kelvin.
- $V_1/T_1 = V_2/T_2$.

MASTER EQUATIONS

$$pV = nRT \leftrightarrow pV = Nk_B T$$

$$N = nN_A \cdot R = N_A k_B$$

Constants: $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ · $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ · $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$

INTERNAL ENERGY

Ideal monatomic gas

- Internal energy is purely kinetic in the ideal-gas model.
- $U = 3/2 nRT$.
- Therefore U depends only on T , not p or V separately.

Change in internal energy

- $\Delta U = 3/2 nR\Delta T$.
- T rises $\rightarrow \Delta U$ positive.
- T falls $\rightarrow \Delta U$ negative.
- T constant $\rightarrow \Delta U = 0$.

FAST DECISION TREE

Before calculating, ask:

1. What stays constant?
2. Are all temperatures in Kelvin?
3. Is the amount of gas fixed?
4. Which proportionality matches the constraint?

Memory: Boyle = pressure-volume opposite. Gay-Lussac = pressure-temperature together. Charles = volume-temperature together.

STUDY METHOD: cover the right side / formula boxes $\rightarrow$ recall from memory $\rightarrow$ check $\rightarrow$ repeat.